

Minecraft Coding Fundamentals

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Packet Use

This printable lesson packet is generated from the paired Markdown guide. The Markdown version remains the clickable, neurodivergent-friendly working copy; this PDF carries the same lesson substance in a formal handout format. Evidence claims in this packet are based on transcript-derived lesson notes and the cleaned transcript files in this workspace.

Quick Scan

- **Grade:** Grade 4
- **Grade Hub:** Notes [../grade-4-Notes.md]
- **Schedule Reference:** Spring2026_TeachingSchedule_Derek.md [../school/Spring2026_TeachingSchedule_Ignite/]
- **Workbook Sheet:** 4th in Teaching_Placement-IGNITE Curriculum_ GCSD25-26.xlsx
- **Lesson Focus:** Students enter Minecraft Education coding worlds, use block code to control the agent, solve challenge levels, and debug why a solution fails.
- **CS Alignment Strength:** Strong CS lesson
- **LaTeX Source:** minecraft-coding-fundamentals.tex [../minecraft-coding-fundamentals.tex]
- **Bib File:** minecraft-coding-fundamentals.bib [../minecraft-coding-fundamentals.bib]
- **Artifacts Folder:** artifacts/ [../artifacts/]

Lesson Identity

- **Likely Class Blocks:** Day 1 – 1:15-2:10 – Baris, Day 2 – 1:15-2:10 – Fowler, Day 3 – 1:15-2:10 – Schrank, Day 4 – 1:15-2:10 – Autore
- **Main Lesson Family:** Minecraft Coding Fundamentals
- **Primary Evidence Base:** Transcript-derived notes plus source transcripts from this teaching-placement workspace.
- **Primary External Source Type:** Official curriculum/provider materials.

What We Are Teaching

- navigating Minecraft Education into the correct coding world
- using sequences, loops, and challenge-specific agent commands
- reading the task objective and translating it into code steps
- debugging timing, movement, and dual-goal challenges

CS Standards Alignment

- **1B-AP-10:** Create programs that include sequences, events, loops, and conditionals. Why it fits here: Students solve levels by assembling block code for the Minecraft agent.
- **1B-AP-11:** Decompose problems into smaller, manageable subproblems. Why it fits here: Students break a level into movement, turning, timing, and agent tasks.
- **1B-AP-15:** Test and debug a program or algorithm to ensure it runs as intended. Why it fits here: Debugging is explicit in the transcripts and in the official challenge structure.

NYS / Local Curriculum Alignment

- Aligned to the local Grade 4 workbook and class sequence for Minecraft-based coding, internet safety, and standards-linked challenge work.
- Supports New York-facing digital fluency and problem-solving goals through structured computational tasks rather than open build time.
- This packet is a core CS lesson family because the transcript evidence shows the teacher explicitly separating “lesson mode” from free building and tying the work to required standards.

Lesson Content and Concepts

- sequence
- loop
- agent
- debugging
- timed challenge
- world navigation

Evidence / Proof of Instruction

- **Transcript-derived note:** 260302 Grade 4 Baris Minecraft Test [../transcript-derived-lessons/260302-grade-4-baris-minecraft-test.md] The note shows the teacher explicitly explaining that Minecraft time should move from building to standards-based coding lessons.
- **Transcript-derived note:** 260313 Grade 4 Baris Minecraft Fundamentals [../transcript-derived-lessons/260313-grade-4-baris-minecraft-fundamentals.md] The lesson note captures the coding-world entry path, agent control, and repeated troubleshooting moves.
- **Transcript-derived note:** 260312 Grade 4 Baris Agent Challenge [../transcript-derived-lessons/260312-grade-4-baris-agent-challenge.md] The challenge note documents timed dual-plate activation, debugging, and loop-based progression.
- **Source transcript:** 260313 Grade 4 Baris Minecraft Fundamentals [../transcripts/260313-grade-4-baris-minecraft-fundamentals.md] The transcript records menu navigation to Computer Science > Block Coding > Upper Primary > Coding Fundamentals 1, along with repeated reminders to read directions and follow the lesson world.
- **Source transcript:** 260302 Grade 4 Baris Minecraft Test [../transcripts/260302-grade-4-baris-minecraft-test.md] The transcript records the teacher saying the goal is to use Minecraft to teach coding, internet safety, and standards, not simply to let students build.
- **Likely student proof:** World-entry success, completed agent challenges, debugged code, screenshots, and level completion all serve as proof that the CS content was taught.

Assessment Evidence

- Student enters the correct world and lesson with reduced support.
- Student uses agent code to solve at least one structured challenge.
- Student can explain a bug and the change that fixed it.
- Student distinguishes lesson-based coding from free build time.

UDL and Inclusion Supports

Multiple Representation

- Project the exact menu path and keep it visible during startup.
- Use one shared visual for command vocabulary such as move, turn, loop, and agent.
- Show the objective and the success condition before students start coding.

Multiple Action / Expression

- Allow students to plan with arrows or pseudocode before placing blocks.
- Use partners strategically so one student navigates while the other checks directions.
- Provide a “help checkpoint” routine: read objective, test one block, ask after trying one fix.

Multiple Engagement

- Use challenge progression and visible level goals to sustain attention.
- Normalize being stuck as part of coding, especially in the boss-level and timed tasks.
- Keep free-build desires acknowledged but separated from the current lesson purpose.

How We Are Already Making It Inclusive

- Transcript evidence shows strong teacher modeling of sign-in and world navigation.
- The teacher repeatedly re-centers students on reading directions, which supports students who otherwise rush into trial-and-error.
- Pairing and quiet-wait expectations are already used to manage support demand and help students stay in the lesson flow.

How to Improve Inclusion Next Time

- Add printed startup cards with screenshots for the login and lesson path.
- Create one low-floor entry task and one stretch task inside the same packet so students do not bottleneck at the same point.
- Add a debug log page students can mark with “what I tried” and “what changed.”
- Save screenshots of completed levels, common bugs, and successful fixes in **artifacts/** as proof of teaching.

Materials and Setup

- Student devices
- Minecraft Education installed and updated
- Projector / modeling screen
- World access or library setup
- Printed startup / debug cards

Teacher Moves / Script / Routines

- Start by naming the lesson purpose and separating it from free build mode.
- Model the exact clicks to reach the lesson world, then pause for students to mirror.
- Require students to read the challenge objective before placing blocks.
- Stop mid-lesson to share one debugging pattern that is working.
- Close with a brief reflection on the challenge, bug, and fix.

Common Errors and Debugging

- Students may get lost in sign-in or library navigation before reaching the lesson.
- Students may ignore written directions and brute-force the level.
- Timed or dual-goal levels can overload students if they do not break the task into parts.
- Some students may confuse avatar movement and agent movement.

Extensions and Simplifications

- Extension: ask students to optimize a working solution with fewer blocks or more elegant loop use.
- Simplification: provide a partially completed block stack or a paper path plan first.
- Extension: add a reflection comparing two different solutions to the same level.
- Simplification: assign a fixed partner role split for navigation, coding, and checking.

Related Local Materials

- Minecraft CS Module Notes [[../../lessons/minecraft_cs_module_warner/Minecraft-CS-Module-Notes.md](#)]

Artifacts Checklist

- Compiled lesson PDF in **artifacts/**
- At least one screenshot, student example, or setup photo
- One short assessment or observation record
- Updated standards/source bibliography once the **.bib** file is revised
- Any teacher reflection or revision notes after reteaching

Selected Official Sources

Selected official sources that informed this packet are listed below. ocite*

References